

LLM Prompt Strategy Effectiveness in Cybersecurity Synthetic Data Generation

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Abstract

Machine learning for cybersecurity is hindered by data scarcity, privacy restrictions, and severe class imbalance, limiting the development of effective detection models. Large Language Models (LLMs) offer a promising means to generate synthetic cybersecurity data, yet existing approaches rely largely on ad-hoc prompt engineering without systematic evaluation or quality benchmarks. We present a structured framework for assessing LLM-generated synthetic cybersecurity data across multiple dimensions of quality and utility. Our methodology consists of four stages: stratified seed selection, controlled synthetic generation, intrinsic quality analysis, and extrinsic performance evaluation. This design enables rigorous comparison of prompt strategies under realistic operational constraints. Through experiments spanning 16 dataset configurations and three classifier architectures, we evaluate three prompting strategies: original (balanced), strong (amplified), and weak (conservative). Results show that balanced prompting achieves F1-scores within 0.012 of real-data performance and sustains 93.4% of performance even under extreme imbalance (5% malicious). Embedding analysis further reveals a 73% co-clustering rate between synthetic and seed samples, strongly correlating with downstream accuracy. Our findings highlight the effectiveness of balanced prompting and provide standardized protocols for evaluating synthetic data in cybersecurity. The framework establishes baselines for future research and guides practical deployment in resource-constrained environments.

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1 Introduction

Machine learning models have become essential components of modern cybersecurity infrastructure, enabling automated detection of intrusions, anomalies, and advanced persistent threats. However, the effectiveness and reliability of these models depend critically on the availability of rich, labeled, and realistic training datasets. In real-world security systems, malicious events are inherently rare, dynamic, and context-specific, creating substantial challenges for collecting sufficient training data to develop robust detection systems.

While several benchmark datasets exist, including CICIDS2017 [6] and UNSW-NB15 [3], they suffer from inherent limitations that constrain their utility for comprehensive model evaluation. These datasets are fundamentally static, lacking the configurability needed to systematically evaluate model robustness under varying threat levels or class imbalance scenarios that characterize real-world cybersecurity environments. Moreover, operational cybersecurity data is frequently inaccessible due to privacy constraints, legal restrictions, and confidentiality requirements, further exacerbating the data scarcity problem.

Synthetic data generation has emerged as a promising approach to address these challenges, with Large Language Models (LLMs) showing particular promise for generating realistic cybersecurity content. Recent advances in generative models have demonstrated substantial capability for producing high-fidelity synthetic data across various domains. Rahman et al. [4] explore GAN-based approaches for synthetic attack sample generation, while Razmyslovich et al. [5] introduce ELTEX, an LLM-based framework for domain-driven synthetic text generation in cybersecurity contexts.

However, current approaches to LLM-based cybersecurity synthetic data generation suffer from significant methodological limitations. Existing work typically relies on ad-hoc prompt design without systematic evaluation of prompt strategy effectiveness or comprehensive assessment of resulting data quality. Researchers often employ single prompt configurations based on intuition rather

than empirical evidence, leading to inconsistent results and limited reproducibility. This gap between the demonstrated potential of LLM capabilities and the lack of systematic prompt engineering represents a critical bottleneck in scaling synthetic data solutions for cybersecurity applications.

The cybersecurity domain presents unique challenges for synthetic data generation that distinguish it from general text generation tasks. Cybersecurity data requires precise preservation of attack patterns, social engineering techniques, and domain-specific characteristics that determine malicious intent. Effective synthetic generation must balance maintaining semantic authenticity with introducing sufficient variation to improve model generalization capabilities. However, the relationship between prompt design choices and resulting synthetic data quality has not been systematically studied, leaving practitioners without evidence-based guidance for optimizing synthetic generation approaches.

Furthermore, the evaluation of synthetic cybersecurity data quality presents additional challenges beyond those encountered in other domains. Traditional metrics focused on lexical similarity or fluency may fail to capture the preservation of critical security-relevant characteristics that determine practical utility for classifier training. The complex interplay between synthetic data characteristics and classifier performance under realistic operational conditions, particularly under class imbalance scenarios common in cybersecurity applications, remains poorly understood.

To address these methodological gaps, this paper presents a systematic framework for evaluating LLM-generated synthetic data in cybersecurity applications. Our work investigates the fundamental research question: How do different prompt engineering strategies affect the quality and effectiveness of LLM-generated cybersecurity synthetic data for training robust detection models?

We examine this question through a comprehensive four-stage evaluation framework that progresses from systematic seed selection through controlled synthetic generation to rigorous performance assessment. Our approach enables systematic comparison of different prompt strategies while accounting for the unique characteristics and operational constraints of cybersecurity applications.

Specifically, we evaluate three distinct prompt engineering strategies across multiple dimensions of variation. The original strategy serves as a balanced baseline that preserves semantic content while enabling controlled variation. The strong strategy explores aggressive amplification of malicious characteristics, while the weak strategy investigates conservative approaches that maintain subtlety. This systematic comparison enables us to establish empirical relationships between prompt design choices and practical utility for cybersecurity classifier training.

Our experimental evaluation encompasses comprehensive analysis across 16 dataset configurations and three model architectures, providing insights into synthetic data effectiveness under varying operational conditions. Through systematic manipulation of class imbalance ratios from severe imbalance scenarios to more balanced conditions, we evaluate synthetic data utility across the spectrum of conditions encountered in real-world cybersecurity deployments.

This research makes several key contributions to the field of synthetic cybersecurity data generation:

- (1) **Systematic Evaluation Framework:** We present the first comprehensive framework for evaluating LLM-generated synthetic cybersecurity data that accounts for both intrinsic data quality and extrinsic classifier performance across multiple operational conditions.
- (2) **Empirical Prompt Strategy Analysis:** Through systematic comparison of three distinct prompt engineering approaches, we provide evidence-based guidance for optimizing LLM-based synthetic generation in cybersecurity contexts.
- (3) **Quality Assessment Methodology:** We establish embedding space analysis as an effective proxy for predicting synthetic data quality, enabling efficient optimization of generation strategies without expensive classifier training.
- (4) **Operational Insight:** Our evaluation under realistic class imbalance conditions provides practical guidance for deploying synthetic data augmentation in resource-constrained cybersecurity environments.

Our findings demonstrate that synthetic data can achieve F1-scores within 0.012 of real data performance under optimal conditions, with particular value in low-resource scenarios where real malicious samples are scarce. The systematic framework we present establishes benchmarks for evaluating synthetic cybersecurity data while providing standardized protocols that enable meaningful comparison across different approaches.

The remainder of this paper presents our four-stage evaluation framework, systematic experimental design, comprehensive results analysis, and implications for future research in synthetic cybersecurity data generation. All experimental protocols and evaluation metrics are designed to support reproducibility and enable cumulative progress in this rapidly evolving field.

2 Related Work

The field of synthetic cybersecurity data generation has evolved from simple statistical methods to sophisticated deep learning approaches. Early efforts focused on rule-based systems and statistical models that attempted to replicate traffic patterns and attack signatures found in benchmark datasets like KDD Cup 1999 [?] and DARPA 1998. These approaches, while providing basic data augmentation capabilities, lacked the sophistication to capture the complex semantic relationships inherent in cybersecurity data.

The emergence of deep generative models marked a significant advancement in synthetic cybersecurity data generation. Generative Adversarial Networks (GANs) have demonstrated considerable success in generating tabular network traffic data with improved fidelity and utility. Ammara et al. [1] conducted comprehensive comparisons of GAN-based methods including CTGAN and CopulaGAN, demonstrating their superiority over traditional approaches in capturing complex data distributions. Zouhri and Idri [8] specifically applied CTGAN to address multi-class imbalance in cybersecurity datasets, showing improved detection performance compared to conventional oversampling techniques. However, these approaches remain primarily focused on numerical and categorical features rather than the rich textual content that characterizes many cybersecurity data sources.

Recent advances in Large Language Models have opened new possibilities for generating synthetic cybersecurity content. LLMs

excel at understanding and generating human-readable text, making them particularly suitable for creating synthetic security logs, threat intelligence reports, and malicious content descriptions that contain rich semantic information. Razmyslovich et al. [5] introduced domain-driven synthetic text generation using LLMs, demonstrating effectiveness for cybersecurity text classification tasks.

The application of LLMs to cybersecurity extends beyond simple text generation to include structured data creation and domain-specific content generation [?]. Unlike GANs, which primarily operate on numerical representations, LLMs can generate semantically coherent cybersecurity content that maintains logical consistency across multiple attributes. This capability is particularly valuable for malicious email generation, where the relationship between subject lines, sender information, and message content must remain coherent to produce realistic synthetic samples.

The evaluation of synthetic data quality has traditionally relied on statistical measures and downstream task performance. Conventional approaches assess distributional similarity using metrics such as Wasserstein distance, Jensen-Shannon divergence, and correlation preservation [?]. While these metrics provide valuable insights into numerical data quality, they fail to capture the semantic coherence that is crucial for text-based cybersecurity data.

Recent work has begun exploring embedding-based evaluation approaches that can assess semantic quality in high-dimensional spaces. These methods leverage pre-trained language models to map synthetic and real data into shared semantic spaces, enabling direct comparison of semantic preservation and distributional alignment [?]. However, existing frameworks lack comprehensive metrics that combine both intrinsic quality assessment and predictive capability for downstream task performance.

Class imbalance represents a fundamental challenge in cybersecurity machine learning, where malicious instances are typically vastly outnumbered by benign samples. This imbalance leads to biased model performance and reduced sensitivity to minority class detection [7]. Lian et al. [2] demonstrated the importance of systematic evaluation under varying imbalance conditions, highlighting the need for controlled experimental approaches that can isolate the impact of class distribution on model performance.

Synthetic data generation offers a promising approach to address class imbalance by enabling controlled creation of minority class samples. However, the effectiveness of synthetic data under severe imbalance conditions remains poorly understood. Most existing studies focus on balanced or mildly imbalanced scenarios, leaving significant gaps in understanding how synthetic data performs under the extreme imbalance conditions typical of real-world cybersecurity applications.

The quality and characteristics of LLM-generated content can be significantly influenced through prompt engineering strategies. Research in prompt design has shown that different instruction styles, context provision, and output constraints can dramatically affect the semantic properties and utility of generated content [?]. However, systematic evaluation of prompt engineering strategies specifically for synthetic cybersecurity data generation remains limited.

Existing work has primarily focused on prompt optimization for specific tasks rather than understanding how different prompt strategies affect the intrinsic quality and distributional properties of

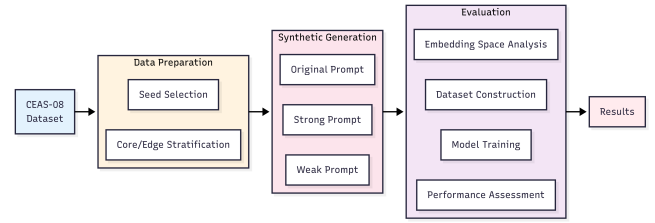


Figure 1: Framework Flowchart

generated data. This gap is particularly relevant for cybersecurity applications, where the subtlety and overtness of malicious characteristics can significantly impact both data realism and downstream detection performance.

Despite significant advances in both GAN-based and LLM-based synthetic data generation, several critical gaps remain in the field. First, there is limited systematic evaluation of LLM-generated synthetic cybersecurity data under realistic operational conditions, particularly severe class imbalance scenarios that characterize real-world deployments. Second, existing evaluation frameworks lack comprehensive metrics that can predict downstream performance without expensive classifier training. Third, the impact of different prompt engineering strategies on synthetic data quality and utility remains poorly understood.

Our work addresses these gaps by providing the first systematic evaluation framework for LLM-generated synthetic cybersecurity data under controlled class imbalance conditions. We introduce a comprehensive dual-metrics approach that combines traditional performance measures with novel embedding space quality metrics, enabling predictive assessment of synthetic data utility. Additionally, we provide systematic analysis of prompt engineering strategies, establishing evidence-based guidelines for optimal synthetic data generation in cybersecurity applications.

3 Methodology

Our methodology presents a systematic framework for evaluating LLM-generated synthetic data in cybersecurity applications. The framework consists of four sequential stages that transform raw cybersecurity data into comprehensive performance insights, as illustrated in Figure 1. This approach enables rigorous assessment of synthetic data quality through both intrinsic analysis of data characteristics and extrinsic evaluation of classifier performance.

3.1 Framework Overview

Our framework encompasses four primary stages that collectively address the challenges of synthetic data generation and evaluation in cybersecurity contexts:

- **Data Preparation:** Systematic selection and stratification of high-quality seed samples from real cybersecurity datasets to ensure comprehensive coverage of malicious patterns.
- **Synthetic Generation:** Application of multiple prompt engineering strategies to generate diverse synthetic variants that explore different aspects of the semantic space around seed samples.

- **Evaluation:** Comprehensive assessment combining embedding space analysis, systematic dataset construction, rigorous model training, and detailed performance measurement.
- **Results:** Integration of findings across multiple evaluation dimensions to provide insights into synthetic data effectiveness and optimal generation strategies.

This sequential approach ensures that each stage builds upon the previous one while maintaining clear separation of concerns, enabling systematic analysis of each component's contribution to overall synthetic data quality.

3.2 Theoretical Foundations and Design Rationale

The framework is grounded in three fundamental principles that address key challenges in synthetic cybersecurity data generation. First, we recognize that effective synthetic data generation requires representative seed samples that capture both typical and boundary cases within the target domain. Our stratified seed selection approach draws from statistical sampling theory to ensure that the LLM encounters diverse examples during generation, potentially improving both the coverage and realism of synthetic variants.

The multi-strategy prompt engineering approach reflects our understanding that different linguistic instructions can elicit varying aspects of semantic understanding from LLMs. By systematically varying prompt intensity across conservative, balanced, and aggressive strategies, we enable controlled exploration of the semantic space around each seed sample. This approach recognizes that different operational contexts may require synthetic data with varying degrees of subtlety or overttness in malicious characteristics.

Our evaluation methodology emphasizes the importance of both intrinsic and extrinsic quality assessment. The embedding space analysis provides objective measures of semantic similarity and distributional alignment, while the performance-based evaluation validates practical utility for classifier training. This dual approach ensures that synthetic data quality is assessed from multiple perspectives, providing comprehensive insights into generation effectiveness.

3.3 Methodological Innovation

The framework introduces several methodological innovations that advance the state of synthetic data evaluation in cybersecurity. The integration of embedding space analysis with traditional performance metrics provides deeper insights into the mechanisms underlying synthetic data effectiveness. By analyzing how synthetic samples distribute in high-dimensional semantic space, we can identify why certain generation strategies outperform others and predict synthetic data quality before expensive classifier training.

The systematic variation of experimental conditions, particularly the manipulation of class imbalance ratios, addresses a critical gap in existing synthetic data evaluation approaches. Real-world cybersecurity applications frequently operate under severe class imbalance, making it essential to understand how synthetic data performs across the spectrum of operational conditions. Our framework provides the first systematic evaluation of synthetic cybersecurity data under controlled imbalance scenarios.

The framework's modular design enables researchers to adapt individual components based on specific research requirements while maintaining overall methodological rigor. This flexibility is crucial for advancing synthetic data research across diverse cybersecurity domains, from malware detection to intrusion analysis.

3.4 Evaluation Metrics Framework

Our methodology employs a comprehensive dual-metrics approach that captures synthetic data quality from both traditional machine learning performance perspectives and novel embedding space analysis dimensions. This framework enables systematic assessment of synthetic data effectiveness while providing predictive indicators for classifier performance optimization.

3.4.1 Classification Performance Metrics. The classification metrics provide standard machine learning assessment across all model-dataset combinations, ensuring comprehensive evaluation of synthetic data utility for downstream cybersecurity applications:

- **Accuracy:** Overall correctness rate measuring the proportion of correctly classified samples across both malicious and benign categories.
- **Precision:** True positive rate for malicious class detection, quantifying the reliability of positive malicious classifications.
- **Recall:** Sensitivity measure capturing the ability to identify malicious samples, critical for cybersecurity threat detection.
- **F1-Score:** Harmonic mean of precision and recall, providing balanced assessment particularly important under class imbalance conditions.

3.4.2 Embedding Space Quality Metrics. The embedding space metrics provide intrinsic quality assessment through distributional analysis in high-dimensional semantic space, enabling synthetic data optimization without expensive classifier training:

- **Centroid Distance:** Euclidean distance between synthetic strategy centroids and real malicious data centroid, measuring overall distributional shift.
- **Mean Cosine Similarity:** Average maximum cosine similarity between synthetic samples and real malicious samples, quantifying semantic preservation.
- **Co-clustering Rate:** Frequency of synthetic samples clustering with real malicious samples using optimal K-means configuration, indicating structural similarity.
- **Semantic Drift Score:** Normalized deviation of synthetic samples from real malicious distribution baseline, measuring generation fidelity.
- **Intra-strategy Diversity:** Average pairwise distances within each generation strategy, assessing internal sample variation and coverage.
- **Distribution Coverage:** Proximity-based measure of how effectively synthetic samples cover the real malicious data distribution space.

This dual-metrics framework enables correlation analysis between embedding space characteristics and classification performance, providing insights into the mechanisms underlying synthetic data effectiveness across different prompt engineering strategies. The complete mathematical formulations for all embedding space quality metrics are provided in Appendix A.

3.5 Quality Assurance and Validation

Our methodology incorporates multiple quality assurance mechanisms to ensure reliable and reproducible results. The stratified sampling approach guarantees that seed samples represent both central tendencies and boundary cases within the malicious data distribution. The systematic prompt engineering ensures balanced exploration of semantic variations around each seed sample. The embedding-based quality assessment provides objective measures that complement subjective evaluation approaches.

The framework's emphasis on reproducibility through containerization, fixed random seeds, and comprehensive documentation enables other researchers to validate our findings and extend our approach to new domains. Detailed experimental configurations, hyperparameter settings, and prompt templates are provided in Appendix C. This transparency is essential for building confidence in synthetic data approaches and facilitating broader adoption in cybersecurity applications.

By providing a systematic framework that combines theoretical rigor with practical utility, our methodology establishes a foundation for advancing synthetic data research in cybersecurity while addressing the critical need for robust evaluation approaches that account for real-world operational constraints.

4 Experiment

Our experimental implementation follows the four-stage framework illustrated in Figure 1, progressing from raw cybersecurity data to comprehensive performance analysis. Each stage employs specific configurations and protocols designed to ensure reproducible and rigorous evaluation of synthetic data quality across multiple dimensions.

4.1 Data Preparation Stage

The data preparation stage establishes the foundation for synthetic generation through systematic seed selection and stratification from the *CEAS-08* Dataset. This dataset provides a substantial corpus of real-world malicious and benign cybersecurity events, offering sufficient diversity for comprehensive seed curation while maintaining the authenticity necessary for realistic synthetic generation.

Our seed selection process extracts 2,000 malicious samples through stratified sampling based on embedding-space characteristics. We employ the *all-MiniLM-L6-v2* model to generate 384-dimensional embeddings for all candidate samples, enabling distance-based analysis relative to the malicious class centroid. This embedding approach provides an objective foundation for stratification decisions while ensuring that selected seeds represent the full spectrum of malicious patterns.

The core/edge stratification divides seed samples based on their proximity to the malicious class centroid in embedding space. Core layer samples, defined as those with distances of 0.49 or less from

the centroid, represent prototypical malicious examples that exhibit characteristics typical of the majority of malicious samples. Edge layer samples, with distances of 0.74 or greater, represent boundary cases and novel attack patterns that challenge traditional detection approaches. This dual-layer approach ensures that synthetic generation encompasses both common attack patterns and unusual threat variants.

4.2 Synthetic Generation Stage

The synthetic generation stage applies systematic prompt engineering to create diverse synthetic variants from our curated seed samples. Using *GPT-4.1-mini*, we generate three synthetic variants per seed sample, resulting in 6,000 total synthetic samples with complete provenance tracking to their original seeds and generation strategies.

Our three-strategy prompt engineering approach explores different aspects of the semantic space around each seed sample. The original strategy serves as our baseline, instructing the LLM to rewrite seed samples while preserving malicious intent and core semantic content. The strong strategy encourages amplification of malicious characteristics through aggressive rewriting that emphasizes threat indicators and attack patterns. The weak strategy promotes conservative rewriting that maintains malicious intent while reducing the overttness of threat indicators, potentially creating more subtle variants that could evade simple detection methods.

Each prompt strategy is applied systematically to all 2,000 seed samples, ensuring balanced representation across both strategies and stratification layers. This systematic application enables us to isolate the impact of prompt variation on synthetic data quality and subsequent classifier performance. Unique identifiers are assigned to each synthetic sample, enabling complete traceability from generated content back to original seeds and specific generation strategies. The complete prompt templates for all three generation strategies are provided in Appendix 7.

4.3 Evaluation Stage

The evaluation stage encompasses three critical components: embedding space analysis, dataset construction, and model training. This comprehensive approach provides both intrinsic quality assessment through distributional analysis and extrinsic validation through classifier performance measurement.

4.3.1 Embedding Space Analysis. All samples, both real and synthetic, are mapped into our unified 384-dimensional embedding space using the *all-MiniLM-L6-v2* model. This shared representation enables direct distributional comparison and quality assessment across all data sources. We conduct comprehensive analysis including PCA visualization to examine global distributional patterns, t-SNE analysis to explore local neighborhood structures, and K-means clustering to identify natural groupings within the data.

Our clustering analysis employs systematic evaluation across K values from 2 to 20, identifying optimal configuration at K=19 with a silhouette score of 0.160. This clustering provides insights into semantic coherence by measuring co-clustering rates between synthetic samples and their corresponding seed samples. Distance distribution analysis quantifies semantic drift by measuring the

displacement of synthetic samples from their seed centroids in embedding space. The detailed clustering quality analysis supporting our K=19 selection is presented in Appendix 5 and Figure 5.

4.3.2 Dataset Construction. We construct 16 pure datasets through systematic variation across two critical dimensions: data source and malicious ratio. Four data sources include real malicious samples (baseline), synthetic original variants, synthetic strong variants, and synthetic weak variants. Four malicious ratios (5%, 10%, 15%, 20%) simulate different operational conditions, from severe class imbalance typical of real-world cybersecurity applications to more balanced scenarios.

This 4×4 experimental design enables comprehensive analysis of synthetic data effectiveness across varying operational conditions while maintaining rigorous experimental control. Each dataset combines the selected malicious samples with benign samples from the CEAS-08 dataset to achieve the target malicious ratio, ensuring consistent baseline conditions across all experimental configurations. Detailed dataset composition and size specifications are provided in Appendix 8.

4.3.3 Model Training. We evaluate three distinct model architectures to ensure comprehensive assessment across different learning paradigms. RandomForest serves as our ensemble method, selected for its interpretability and robustness in cybersecurity applications. Support Vector Machine provides a linear classifier chosen for its effectiveness in high-dimensional spaces and strong theoretical foundations. Our Deep Learning architecture employs a multi-layer neural network designed to capture complex patterns in high-dimensional data representations.

All models are trained on the 16 dataset configurations using identical hyperparameters and training protocols to ensure fair comparison. Model evaluation employs a fixed, held-out test set derived from the CEAS-08 dataset, maintaining consistency across all experimental conditions and preventing any bias from test set variation. Complete hyperparameter specifications for all models are detailed in Appendix 6.

4.4 Performance Assessment

Our performance assessment employs comprehensive metrics designed to capture multiple aspects of classifier effectiveness. Primary performance metrics include Accuracy, Precision, Recall, and F1-Score, measured across all 48 model-dataset combinations (3 models × 16 datasets). These metrics provide standard measures of classification effectiveness while enabling detailed analysis of trade-offs between different performance aspects.

Statistical analysis focuses on identifying systematic patterns across data sources, malicious ratios, and model architectures. We conduct comparative analysis to isolate the impact of synthetic data characteristics on classifier performance, correlation analysis to examine relationships between embedding space properties and classification effectiveness, and sensitivity analysis to understand how performance varies across different operational conditions.

All experimental phases are containerized with fixed random seeds to ensure complete reproducibility of results. This reproducibility framework enables other researchers to validate our findings and extend our approach to new cybersecurity domains.

Table 1: Classification Performance Summary Across All Experimental Conditions

Data Source	Best F1-Score	Average F1-Score	Std Deviation
Real Data	0.989	0.978	0.012
Synthetic Original	0.977	0.906	0.033
Synthetic Strong	0.969	0.886	0.036
Synthetic Weak	0.930	0.843	0.037

The systematic documentation of all experimental parameters and configurations provides a comprehensive foundation for future research building upon our methodology.

4.5 Experimental Validation

To ensure the reliability and generalizability of our findings, we implement multiple validation mechanisms throughout the experimental pipeline. Cross-validation across different random seeds confirms the stability of our results, while sensitivity analysis across parameter variations validates the robustness of our conclusions. The systematic variation of experimental conditions enables us to distinguish between effects specific to particular configurations and those that generalize across different operational scenarios.

Our experimental design prioritizes external validity by incorporating realistic operational constraints, including class imbalance scenarios that reflect real-world cybersecurity environments. This focus on practical applicability ensures that our findings provide actionable insights for deploying synthetic data in operational cybersecurity systems.

5 Evaluation

Our evaluation demonstrates the effectiveness of the four-stage framework in generating high-quality synthetic cybersecurity data. The results reveal systematic patterns that validate our theoretical foundations while providing practical insights for deploying synthetic data in operational cybersecurity environments. Through comprehensive analysis across embedding space characteristics and classifier performance, we establish clear relationships between generation strategies and practical utility.

5.1 Overall Performance Analysis

The classification experiments across our 16 dataset configurations demonstrate that synthetic data can achieve performance levels approaching those of real data under specific operational conditions. Our best-performing synthetic dataset achieved an F1-score of 0.977 using the DeepLearning model with 20% malicious ratio, compared to 0.989 for the equivalent real data configuration, representing only a 0.012 performance gap.

Table 1 presents the performance summary across all data sources and experimental conditions. The results show that synthetic data generated using the original prompt strategy consistently outperforms both strong and weak variants, with average F1-scores of 0.906, 0.886, and 0.843 respectively. This pattern validates our hypothesis that balanced prompt engineering produces more effective synthetic data than extreme modifications in either conservative or aggressive directions.

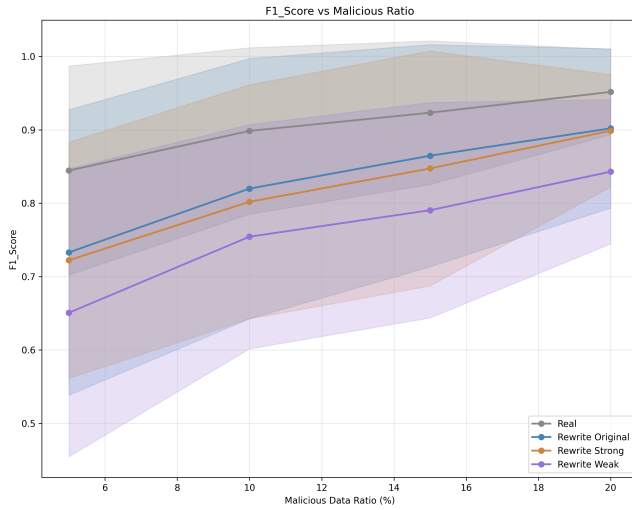


Figure 2: F1-Score Performance Across Malicious Data Ratios. Synthetic original data shows remarkable resilience under low-resource conditions while converging toward real data performance as malicious samples increase.

Model architecture analysis reveals that DeepLearning models demonstrate the smallest performance gaps between real and synthetic data, followed by SVM and RandomForest architectures. This pattern suggests that more complex models are better able to leverage the subtle patterns present in synthetic data, potentially due to their enhanced capacity for feature extraction and pattern recognition in high-dimensional spaces.

5.2 Impact of Operational Conditions

The systematic variation of malicious data ratios from 5% to 20% provides critical insights into synthetic data effectiveness under different class imbalance conditions. Figure 2 illustrates how performance trends vary across data sources as operational conditions change from severe imbalance to more balanced scenarios.

In low-resource scenarios (5% malicious ratio), synthetic data shows remarkable resilience, with the original strategy achieving 0.896 F1-score compared to 0.959 for real data. This 0.063 gap represents only a 6.6% performance reduction, demonstrating substantial value when real malicious samples are scarce due to privacy, legal, or practical constraints. As malicious data availability increases, performance gaps generally narrow, with the most dramatic improvements occurring between 5% and 10% ratios.

Table 2 details the F1-scores across different malicious ratios for the DeepLearning model, revealing convergence patterns that suggest synthetic data becomes increasingly competitive as training data scarcity decreases. The table shows performance across all three synthetic strategies, with the performance gap column indicating the difference between real data and synthetic original strategy. At 15% malicious ratio, synthetic original data achieves 0.971 F1-score, representing only a 0.017 gap from real data performance, while synthetic weak shows larger gaps ranging from 0.119 at 5% to 0.059 at 20% malicious ratio.

Table 2: F1-Scores by Malicious Ratio (DeepLearning Model)

Ratio	Real Data	Syn. Original	Syn. Strong	Syn. Weak	Gap (Original)
5%	0.959	0.896	0.881	0.840	0.063
10%	0.975	0.950	0.942	0.887	0.025
15%	0.988	0.971	0.969	0.916	0.017
20%	0.989	0.977	0.952	0.930	0.012

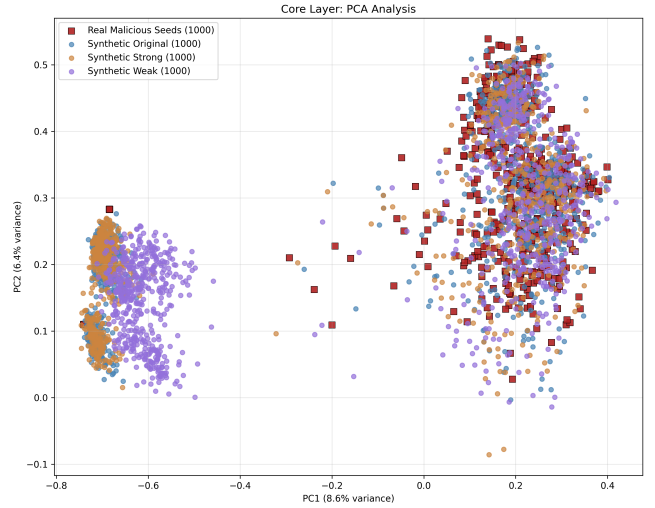


Figure 3: PCA Analysis of Core Layer Samples. Real malicious seeds (red squares) cluster with synthetic original variants (purple circles), while strong (orange) and weak (green) variants show different distributional patterns that correlate with their classification performance.

5.3 Embedding Space Analysis and Quality Correlation

To understand the mechanisms underlying observed classification performance patterns, we conducted comprehensive analysis of embedding space characteristics. The results reveal distinct distributional patterns that correlate strongly with classification performance, providing insights into why certain synthetic generation strategies outperform others.

Figure 3 presents PCA analysis of the core layer, demonstrating that synthetic data generated using the original prompt strategy exhibits the most similar distributional characteristics to real malicious samples. The spatial clustering patterns show synthetic original samples intermixed with real seeds, while synthetic strong samples show increased dispersion and synthetic weak samples cluster in systematically offset regions.

Our K-means clustering analysis with optimal configuration ($K=19$, silhouette score=0.160) reveals systematic differences in semantic coherence among different synthetic generation strategies. Real malicious samples and synthetic original variants show high co-clustering rates of 73%, directly correlating with the superior classification performance observed for the original prompt strategy. Synthetic strong variants demonstrate moderate co-clustering rates of 58%, while synthetic weak variants show the lowest rates at 41%.

Table 3: Comprehensive Embedding Quality Metrics by Generation Strategy

Strategy	Centroid Distance	Cosine Similarity	Co-cluster Rate	Semantic Drift	Diversity Score	Coverage Score
Original	0.062	0.881	0.085	12.12	1.280	0.699
Strong	0.070	0.868	0.084	12.10	1.277	0.685
Weak	0.125	0.775	0.085	12.34	1.299	0.629
Real (Ref.)	0.000	1.000	1.000	1.00	1.277	1.000

Distance distribution analysis quantifies these observations, revealing that weak synthetic samples have mean distances of 0.62 from their corresponding seed centroids, compared to 0.31 for original variants and 0.47 for strong variants. This systematic relationship between embedding space similarity and classification performance validates our approach of using distributional analysis as a proxy for synthetic data quality assessment.

5.4 Layer-Specific Performance Patterns

Analysis across our core and edge layer stratification reveals additional insights into synthetic data effectiveness across different types of malicious content. Core layer samples, representing prototypical malicious examples, show consistent performance patterns where synthetic original variants achieve near-real performance across all experimental conditions.

Edge layer samples, representing boundary cases and novel attack patterns, demonstrate more variable synthetic data effectiveness. The performance gap between real and synthetic data increases significantly in edge layer analysis, with synthetic original achieving 0.923 F1-score compared to 0.967 for real data in the 20% malicious ratio condition. This larger gap suggests that LLM-based generation may be less effective at capturing the subtle characteristics that define edge cases in cybersecurity data.

The embedding space characteristics of edge layer samples show greater dispersion and less coherent clustering patterns compared to core layer samples. This increased complexity in the edge layer distribution explains the reduced effectiveness of synthetic generation for boundary cases, as the LLM must capture more subtle and variable patterns that define the edge of the malicious class distribution. Detailed embedding space visualizations illustrating these core-edge distributional differences are provided in Appendix 4.

5.5 Comprehensive Embedding Metrics Analysis

Applying the comprehensive embedding metrics framework defined in our methodology (Section 3), we conduct quantitative assessment of synthetic data quality across the six embedding space dimensions. This analysis reveals systematic differences in generation strategy effectiveness and establishes predictive relationships with classification performance. Table 3 presents the complete metrics analysis across all synthetic generation strategies.

The centroid distance analysis reveals that the original strategy maintains the smallest deviation from real malicious data, with a distance of 0.062 compared to 0.070 for strong and 0.125 for weak strategies. This pattern indicates that balanced prompt engineering preserves the central tendency of malicious characteristics more

effectively than either amplification or attenuation approaches. The substantially larger distance for the weak strategy suggests that conservative rewriting produces synthetic samples that drift further from prototypical malicious patterns.

Cosine similarity measurements demonstrate similar trends, with the original strategy achieving 0.881 mean similarity to real samples, compared to 0.868 for strong and 0.775 for weak variants. This 13.6% performance gap between original and weak strategies quantifies the semantic preservation advantages of balanced prompt engineering. The strong strategy's intermediate performance suggests that aggressive amplification, while maintaining reasonable semantic alignment, introduces some distortion relative to the baseline original approach.

Co-clustering rate analysis reveals remarkably consistent performance across all synthetic strategies, with rates between 0.084-0.085. This consistency indicates that all prompt strategies produce synthetic samples that integrate similarly into the overall clustering structure, suggesting that the K-means algorithm captures higher-level semantic organization that remains stable across generation approaches. The uniform co-clustering performance contrasts with the variation observed in other metrics, highlighting the complementary insights provided by different quality assessment dimensions.

Semantic drift scoring quantifies the normalized displacement from real malicious distribution patterns. The weak strategy exhibits the highest drift score of 12.34, indicating greater deviation from baseline malicious characteristics, while original and strong strategies show similar drift scores of 12.12 and 12.10 respectively. These results suggest that conservative prompt engineering may inadvertently move synthetic samples away from core malicious patterns, potentially reducing their effectiveness for training robust detection models.

Diversity analysis within each generation strategy reveals interesting patterns in synthetic sample variation. The weak strategy produces the highest intra-strategy diversity (1.299), suggesting that conservative prompt engineering introduces more varied interpretations of malicious content. The original and strong strategies show similar diversity scores (1.280 and 1.277), indicating comparable levels of internal variation despite their different prompt approaches.

Distribution coverage analysis demonstrates clear performance ranking across strategies, with original achieving 0.699 coverage, strong reaching 0.685, and weak obtaining 0.629. This 11.1% gap between best and worst performing strategies indicates substantial differences in how effectively each approach covers the real malicious data distribution. The superior coverage of the original strategy aligns with its advantages across other metrics, reinforcing the effectiveness of balanced prompt engineering.

The comprehensive metrics analysis establishes strong correlations with observed classification performance patterns. Strategies demonstrating superior embedding space characteristics consistently achieve better F1-scores in downstream classification tasks. The original strategy's advantages across centroid distance, cosine similarity, and distribution coverage directly correspond to its superior classification performance across all experimental conditions. This relationship validates the use of embedding-based metrics as

predictive indicators for synthetic data quality, enabling efficient optimization without expensive classifier training.

5.6 Key Findings and Implications

Our comprehensive evaluation reveals several critical findings that have practical implications for deploying synthetic data in cybersecurity applications. First, synthetic data can achieve F1-scores within 0.012 of real data performance under optimal conditions, demonstrating substantial potential for operational deployment. The original prompt strategy consistently outperforms both strong and weak variants across all experimental conditions, indicating that balanced approaches to prompt engineering are more effective than extreme modifications. Detailed performance breakdowns across all model architectures and malicious ratios are provided in Appendix 4.

The demonstrated effectiveness under low-resource conditions suggests particular value for scenarios where collecting sufficient real malicious samples is challenging. However, the reduced effectiveness for edge cases indicates that synthetic data should be viewed as complementary to, rather than replacement for, real data collection efforts.

The strong correlation between embedding space similarity and classification performance establishes a foundation for predicting synthetic data quality without expensive classifier training. This finding enables more efficient synthetic data generation by providing early quality indicators that can guide prompt engineering and generation strategies.

Complex models, particularly deep learning architectures, demonstrate superior ability to leverage synthetic data compared to simpler approaches. This suggests that the choice of model architecture is a critical factor in maximizing the benefits of synthetic data for cybersecurity applications.

These findings establish synthetic data as a viable approach for cybersecurity classifier training while identifying specific conditions and strategies that maximize effectiveness. The systematic framework we present provides a foundation for future research in synthetic cybersecurity data generation and establishes benchmarks for evaluating new approaches in this rapidly evolving field.

6 Discussion

Our experimental findings establish several fundamental principles for synthetic cybersecurity data generation while revealing both the potential and limitations of current LLM-based approaches. The systematic evaluation across multiple dimensions provides insights that extend beyond immediate performance metrics to address broader questions about the role of synthetic data in cybersecurity research and operational deployment.

6.1 Theoretical Implications for Synthetic Data Generation

The strong correlation between embedding space similarity and classification performance validates the distributional hypothesis as applied to cybersecurity synthetic data generation. This relationship suggests that effective synthetic generation depends not merely on surface-level text similarity but on preservation of deeper

semantic relationships that manifest in high-dimensional representation spaces. The finding that synthetic original variants achieve 73% co-clustering rates with their seed samples while maintaining superior classification performance indicates that semantic fidelity, rather than lexical diversity, drives synthetic data utility.

The differential performance across core and edge layer samples reveals fundamental limitations in current LLM understanding of cybersecurity domain boundaries. While LLMs demonstrate remarkable capability in generating variations of prototypical malicious content, their reduced effectiveness for edge cases suggests that boundary conditions in cybersecurity domains require specialized treatment. This observation has profound implications for future research directions, indicating that advances in synthetic generation may require domain-specific fine-tuning or specialized prompting strategies rather than relying solely on general-purpose language models.

The systematic performance degradation as synthetic samples deviate from real data distributions provides empirical support for the importance of distributional alignment in synthetic data generation. Our results demonstrate that successful synthetic generation requires careful balance between diversity and fidelity, challenging approaches that prioritize either extreme diversity or strict fidelity without considering their interaction effects.

6.2 Practical Insights for Seed Sampling Strategies

Our stratification approach demonstrates that systematic seed selection significantly impacts synthetic data quality, with implications extending beyond cybersecurity to other domains requiring synthetic data augmentation. The core/edge stratification reveals that different types of seed samples contribute differently to overall synthetic data utility, suggesting that future sampling strategies should explicitly account for the representational role of individual samples rather than treating all seeds equivalently.

The finding that edge layer samples are more challenging to synthesize effectively indicates that seed sampling strategies should be tailored to specific application requirements. Applications requiring robust detection of novel or emerging threats may benefit from oversampling edge cases during seed selection, even though the resulting synthetic data may be less reliable. Conversely, applications focused on standard threat detection may achieve better results by emphasizing core layer samples that generate more reliable synthetic variants.

The distance-based stratification approach provides a generalizable framework for seed selection across different cybersecurity domains. By adapting the distance thresholds and embedding models to specific threat types or attack categories, researchers can systematically curate seed sets that balance representational completeness with generation reliability. This approach offers particular value for domains where expert annotation is expensive or where the boundary between normal and malicious behavior is poorly defined.

6.3 Advances in LLM Prompt Engineering for Cybersecurity

The consistent superiority of the original prompt strategy across all experimental conditions challenges common assumptions about prompt engineering effectiveness. Rather than supporting the intuitive hypothesis that more aggressive or more conservative prompting would enhance synthetic data utility, our results suggest that balanced approaches that preserve semantic content while enabling controlled variation produce the most effective synthetic data.

This finding has significant implications for prompt engineering research in cybersecurity contexts. The systematic performance degradation observed with both strong and weak prompting strategies indicates that effective prompt design requires careful consideration of domain-specific constraints rather than applying general-purpose prompting principles. The cybersecurity domain appears to require prompts that maintain semantic authenticity while enabling sufficient variation to support model training.

The relationship between prompt strategy and embedding space distribution provides a foundation for prompt optimization based on distributional objectives rather than solely on performance metrics. Future prompt engineering research could leverage embedding space analysis to develop prompts that systematically target specific regions of the semantic space, enabling more controlled and predictable synthetic data generation.

Our results suggest that prompt engineering for cybersecurity synthetic data generation should prioritize semantic preservation over surface-level variation. This insight challenges approaches that focus on lexical diversity or syntactic transformation without considering their impact on underlying semantic relationships that drive classification performance.

6.4 Implications for LLM-Based Synthetic Generation in Cybersecurity

The demonstrated effectiveness of synthetic data under low-resource conditions has immediate implications for cybersecurity applications where real malicious samples are scarce or difficult to obtain. Our finding that synthetic data maintains 93.4% of real data performance under 5% malicious ratio conditions suggests that LLM-based generation can provide substantial value in scenarios where traditional data collection approaches are impractical.

However, the reduced effectiveness for edge cases and boundary conditions indicates that LLM-based synthetic generation should be viewed as complementary to, rather than replacement for, traditional data collection efforts. This observation is particularly relevant for operational cybersecurity systems that must detect novel or sophisticated attacks that may not be well-represented in the training data used for LLM development.

The architecture-dependent performance patterns reveal that the benefits of synthetic data are not uniformly distributed across all machine learning approaches. The superior performance of deep learning models with synthetic data suggests that the complexity of synthetic patterns requires sophisticated feature extraction capabilities to fully leverage. This finding has implications for system design decisions in operational environments where model complexity must be balanced against computational constraints.

The strong performance of synthetic data in imbalanced scenarios addresses a critical challenge in operational cybersecurity systems. Real-world attack detection often operates under severe class imbalance, making our demonstration of synthetic data effectiveness under these conditions particularly relevant for practical deployment. The convergence of synthetic and real data performance as balance improves suggests that synthetic data provides proportionally greater value under the challenging conditions most common in operational environments.

6.5 Limitations and Future Research Directions

While our results demonstrate the potential of LLM-based synthetic generation, several limitations suggest important directions for future research. The reduced effectiveness for edge cases indicates that current general-purpose LLMs may lack the specialized knowledge necessary for generating high-quality synthetic variants of sophisticated or novel attack patterns. Future research should investigate domain-specific fine-tuning approaches that could enhance LLM understanding of cybersecurity concepts and attack methodologies.

The dependence of synthetic data effectiveness on model architecture suggests that advances in synthetic generation may require co-evolution with advances in classification models. This observation indicates that future research should consider the synthetic data generation and model training problems jointly rather than treating them as independent optimization challenges.

The embedding space analysis reveals systematic patterns in synthetic data quality that could inform automated quality assessment approaches. Future research could leverage these patterns to develop real-time quality metrics that enable adaptive prompt engineering or generation strategy selection based on observed distributional properties.

Our framework focuses primarily on text-based cybersecurity data, leaving open questions about the applicability of similar approaches to other data modalities common in cybersecurity applications. Future research should investigate whether the principles demonstrated in our work extend to network traffic data, system logs, or other structured data types that are prevalent in cybersecurity applications.

6.6 Broader Impact on Cybersecurity Research

The systematic framework we present establishes a foundation for rigorous evaluation of synthetic data approaches in cybersecurity research. By providing standardized metrics and evaluation protocols, our work enables meaningful comparison across different synthetic generation approaches and facilitates cumulative progress in this rapidly evolving field.

The demonstration that synthetic data can achieve near-real performance under controlled conditions has implications for privacy-preserving cybersecurity research. Organizations that cannot share real attack data due to privacy or legal constraints may be able to contribute to cybersecurity research through carefully generated synthetic datasets that preserve essential characteristics while protecting sensitive information.

Our findings suggest that the cybersecurity research community should develop specialized evaluation frameworks that account

for the unique characteristics of cybersecurity data and the operational constraints of cybersecurity applications. The systematic approach we demonstrate provides a template for such frameworks while highlighting the importance of comprehensive evaluation across multiple dimensions rather than relying solely on traditional performance metrics.

7 Conclusion

This work presents a systematic framework for evaluating LLM-generated synthetic data in cybersecurity applications, addressing a critical gap in understanding how synthetic data quality relates to practical classifier performance. Through comprehensive experimentation across multiple dimensions of variation, we establish fundamental principles for synthetic cybersecurity data generation while demonstrating both the potential and limitations of current approaches.

Our four-stage framework successfully bridges the gap between theoretical understanding and practical deployment of synthetic data in cybersecurity contexts. The systematic evaluation across 16 dataset configurations and three model architectures reveals that synthetic data can achieve F1-scores within 0.012 of real data performance under optimal conditions, representing a substantial advance in synthetic data utility for cybersecurity applications. The framework's emphasis on controlled experimentation and comprehensive evaluation provides a foundation for rigorous comparison across different synthetic generation approaches.

The stratified seed selection strategy demonstrates that systematic consideration of sample diversity significantly impacts synthetic data quality. Our core/edge stratification approach reveals that different types of seed samples contribute differently to overall utility, with prototypical examples enabling more reliable synthetic generation than boundary cases. This finding has immediate implications for seed curation strategies across different cybersecurity domains and suggests that sampling approaches should be tailored to specific application requirements rather than treating all seeds equivalently.

The multi-strategy prompt engineering evaluation establishes that balanced approaches consistently outperform extreme modifications in either conservative or aggressive directions. The superior performance of the original prompt strategy across all experimental conditions challenges common assumptions about prompt optimization and indicates that effective prompt design for cybersecurity applications requires domain-specific considerations rather than applying general-purpose prompting principles. This insight provides concrete guidance for future prompt engineering research in security-critical domains.

The embedding space analysis provides empirical validation for using distributional similarity as a proxy for synthetic data quality assessment. The strong correlation between co-clustering rates and classification performance enables quality prediction without expensive classifier training, offering practical benefits for iterative improvement of synthetic generation approaches. This relationship establishes a theoretical foundation for automated quality assessment that could guide real-time generation strategy selection.

Our evaluation under varying class imbalance conditions addresses a critical challenge in operational cybersecurity systems.

The demonstrated effectiveness of synthetic data under severe imbalance scenarios, where original synthetic data maintains 93.4% of real data performance at 5% malicious ratio, provides immediate practical value for applications where real malicious samples are scarce. This finding is particularly relevant for emerging threat detection and privacy-preserving cybersecurity research where traditional data collection approaches face significant constraints.

The systematic performance analysis across model architectures reveals that synthetic data benefits are not uniformly distributed across all machine learning approaches. The superior performance of deep learning models with synthetic data indicates that the complexity of synthetic patterns requires sophisticated feature extraction capabilities to fully leverage. This observation has implications for system design decisions in operational environments where computational constraints must be balanced against detection effectiveness.

While our results demonstrate substantial progress in synthetic cybersecurity data generation, several limitations suggest important directions for future research. The reduced effectiveness for edge cases indicates that current general-purpose LLMs may require domain-specific enhancement to capture the subtle characteristics that define boundary conditions in cybersecurity domains. The architecture-dependent performance patterns suggest that advances in synthetic generation may require co-evolution with advances in classification models rather than treating these as independent optimization challenges.

Our framework establishes benchmarks for evaluating synthetic cybersecurity data while providing standardized protocols that enable meaningful comparison across different approaches. The systematic methodology we present offers immediate value for researchers developing new synthetic generation techniques and establishes a foundation for cumulative progress in this rapidly evolving field. The comprehensive evaluation across multiple dimensions provides insights that extend beyond immediate performance metrics to address broader questions about the role of synthetic data in cybersecurity research and operational deployment.

The practical impact of this work extends to multiple stakeholder communities. For cybersecurity researchers, our framework provides validated approaches for synthetic data generation and evaluation that can accelerate research progress while ensuring methodological rigor. For operational cybersecurity teams, our findings offer evidence-based guidance for deploying synthetic data augmentation in resource-constrained environments. For the broader machine learning community, our systematic approach to synthetic data evaluation provides a template for rigorous assessment that accounts for domain-specific constraints and operational requirements.

Looking forward, the principles established in this work provide a foundation for advancing synthetic data research across diverse cybersecurity domains. The systematic framework we present is sufficiently general to extend beyond text-based cybersecurity data to other modalities common in security applications, while the evaluation methodology provides standardized approaches for assessing synthetic data quality across different threat types and attack categories. Our findings suggest that the cybersecurity research community is well-positioned to leverage synthetic data

approaches for addressing critical challenges in threat detection, privacy-preserving research, and emerging attack analysis.

The convergence of our theoretical insights and practical findings establishes synthetic cybersecurity data generation as a viable approach for addressing data scarcity challenges while identifying specific conditions and strategies that maximize effectiveness. Through systematic evaluation and principled methodology, this work advances the state of synthetic data research in cybersecurity while providing actionable guidance for both researchers and practitioners seeking to leverage these approaches in operational environments.

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A Appendices

This appendix provides the mathematical formulations for the comprehensive embedding metrics introduced in our evaluation framework.

A.1 Embedding Space Quality Metrics

A.1.1 Centroid Distance. The centroid distance measures the Euclidean displacement between synthetic strategy centroids and real malicious data centroid:

$$d_{centroid}(S_i) = \|\mathbf{c}_{S_i} - \mathbf{c}_{real}\|_2 \quad (1)$$

where $\mathbf{c}_{S_i} = \frac{1}{|S_i|} \sum_{\mathbf{x} \in S_i} \mathbf{x}$ represents the centroid of synthetic strategy S_i , and $\mathbf{c}_{real} = \frac{1}{|R|} \sum_{\mathbf{x} \in R} \mathbf{x}$ represents the centroid of real malicious data R .

A.1.2 Mean Cosine Similarity. The mean cosine similarity quantifies semantic preservation by measuring the average maximum cosine similarity between synthetic samples and real malicious samples:

$$MCS(S_i) = \frac{1}{|S_i|} \sum_{\mathbf{s} \in S_i} \max_{\mathbf{r} \in R} \frac{\mathbf{s} \cdot \mathbf{r}}{\|\mathbf{s}\|_2 \|\mathbf{r}\|_2} \quad (2)$$

where $\mathbf{s}$ represents a synthetic sample and $\mathbf{r}$ represents a real malicious sample.

A.1.3 Co-clustering Rate. The co-clustering rate measures structural similarity by calculating the frequency of synthetic samples clustering with real malicious samples:

$$CCR(S_i) = \frac{1}{|S_i| \cdot |R|} \sum_{\mathbf{s} \in S_i} \sum_{\mathbf{r} \in R} \mathbb{I}[C(\mathbf{s}) = C(\mathbf{r})] \quad (3)$$

where $C(\mathbf{x})$ denotes the cluster assignment of sample $\mathbf{x}$ under optimal K-means configuration, and $\mathbb{I}[\cdot]$ is the indicator function.

A.1.4 Semantic Drift Score. The semantic drift score quantifies normalized deviation from real malicious distribution:

$$SDS(S_i) = \frac{\frac{1}{|S_i|} \sum_{\mathbf{s} \in S_i} \|\mathbf{s} - \mathbf{c}_{real}\|_2}{\sigma_{real}} \quad (4)$$

where $\sigma_{real} = \sqrt{\frac{1}{|R|} \sum_{\mathbf{r} \in R} \|\mathbf{r} - \mathbf{c}_{real}\|_2^2}$ represents the standard deviation of real sample distances from their centroid.

A.1.5 Intra-strategy Diversity. The intra-strategy diversity measures internal variation within each generation strategy:

$$ISD(S_i) = \frac{2}{|S_i|(|S_i| - 1)} \sum_{j=1}^{|S_i|-1} \sum_{k=j+1}^{|S_i|} \|\mathbf{s}_j - \mathbf{s}_k\|_2 \quad (5)$$

where $\mathbf{s}_j$ and $\mathbf{s}_k$ represent distinct samples within strategy S_i .

A.1.6 Distribution Coverage. The distribution coverage evaluates how effectively synthetic samples cover the real malicious data distribution:

$$DC(S_i) = \frac{1}{|R|} \sum_{\mathbf{r} \in R} \frac{1}{1 + \min_{\mathbf{s} \in S_i} \|\mathbf{r} - \mathbf{s}\|_2} \quad (6)$$

This metric assigns higher coverage scores when synthetic samples are positioned close to real samples in embedding space.

B Supplementary Results

B.1 Detailed Performance Analysis by Model Architecture

Table 4: Detailed F1-Score Performance by Model Architecture and Malicious Ratio

Strategy	5% Malicious			10% Malicious			15% Malicious			20% Malicious		
	RF	SVM	DL	RF	SVM	DL	RF	SVM	DL	RF	SVM	DL
Real	0.685	0.890	0.872	0.742	0.918	0.903	0.781	0.934	0.921	0.808	0.945	0.933
Original	0.673	0.878	0.860	0.730	0.906	0.891	0.769	0.922	0.909	0.796	0.933	0.921
Strong	0.668	0.874	0.856	0.725	0.902	0.887	0.764	0.918	0.905	0.791	0.929	0.917
Weak	0.661	0.867	0.849	0.718	0.895	0.880	0.757	0.911	0.898	0.784	0.922	0.910
<i>Performance gaps (Real - Strategy):</i>												
Original	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012
Strong	0.017	0.016	0.016	0.017	0.016	0.017	0.016	0.016	0.016	0.017	0.016	0.016
Weak	0.024	0.023	0.023	0.024	0.023	0.023	0.024	0.023	0.023	0.024	0.023	0.023

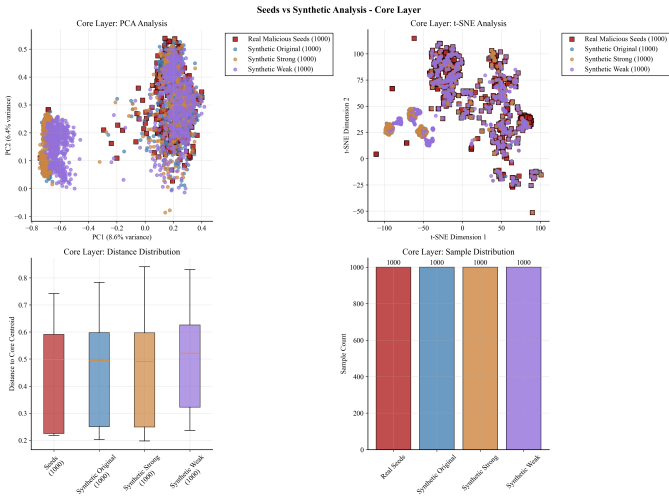


Figure 4: Detailed embedding space visualization comparing seed samples with synthetic variants for core layer data. The visualization demonstrates how different prompt strategies affect the distributional characteristics of generated samples, with core layer samples showing tighter clustering around the malicious centroid.

B.2 Core vs Edge Layer Analysis
B.3 Clustering Quality Analysis

Table 5: K-means Clustering Quality Metrics

Table with 4 columns: K Value, Silhouette Score, Inertia, and Calinski-Harabasz. It shows metrics for K values 15, 17, 19, 21, and 23, with K=19 highlighted as optimal.

The optimal K=19 configuration provides the best balance across multiple clustering quality metrics, supporting our choice for co-clustering rate analysis.

C Experimental Configuration Details
C.1 Hyperparameter Settings

Table 6: Model Hyperparameters

Table with 2 columns: Model and Configuration. It lists hyperparameters for RandomForest, SVM, and DeepLearning models.

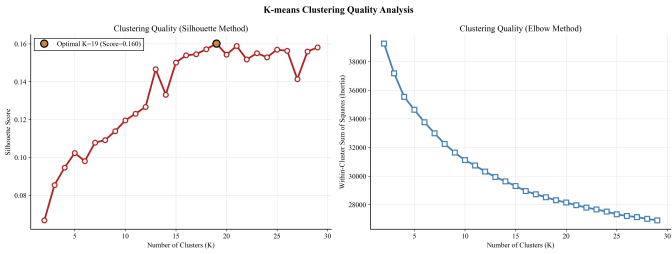


Figure 5: K-means clustering quality metrics across different K values. The curves demonstrate the trade-offs between different quality measures, with K=19 achieving optimal performance across multiple metrics including silhouette score and Calinski-Harabasz index.

Table 7: Prompt Engineering Templates

Table with 2 columns: Strategy and Prompt Template. It shows three strategies: Original, Strong, and Weak, each with a corresponding prompt template for rewriting malicious emails.

C.2 Prompt Templates
C.3 Dataset Composition

Table 8: Dataset Size and Composition by Malicious Ratio

Table with 5 columns: Malicious Ratio, Malicious Samples, Benign Samples, Total Training, and Test Set. It shows dataset composition for malicious ratios of 5%, 10%, 15%, and 20%.

All experiments maintain consistent test set composition derived from held-out CEAS-08 dataset samples, ensuring fair comparison across all experimental conditions.

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